

Course Syllabus

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EE 102: Signal Processing and Linear Systems

Instructor: Ayush Pandey (meet with me 1-on-1 using this [URL](https://outlook.office.com/bookwithme/user/01b63bdd7fca46e09e7832292840538e@ucmerced.edu/meetingtype/iseQLcF3dkS9_VQh-6DdLw2?anonymous&ep=mink)  [_](https://outlook.office.com/bookwithme/user/01b63bdd7fca46e09e7832292840538e@ucmerced.edu/meetingtype/iseQLcF3dkS9_VQh-6DdLw2?anonymous&ep=mink); course [GitHub](https://github.com/ee-ucmerced/ee102-signals-systems)  [_](https://github.com/ee-ucmerced/ee102-signals-systems))

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 **Class Discussions:** Mon and Wed from 1.30pm to 3.20pm at COB1 120

 **Ayush's office hours:** Wednesdays at 3.30pm in SE2 381

 **Lab (Discussion Section):**

- 02D: Thursday 4:30 PM - 7:20 PM in COB1 114
- 03D: Friday 1:30 PM - 4:20 PM in COB1 276
- 04D: Friday 10:30 AM - 1:30 PM in COB1 282

 **GitHub:** <https://github.com/ee-ucmerced/ee102-signals-systems>  [_](https://github.com/ee-ucmerced/ee102-signals-systems)

Textbooks and Resources:

1. Oppenheim, A. V., Willsky, A. S., Nawab, S. H. (1997). "Signals & Systems". Prentice Hall. ISBN: 9780136511755, 0136511759
 - A. Freely available lecture notes, videos, and problem sets are available on MIT's Open Course Ware (OCW) webpage [here](https://ocw.mit.edu/courses/res-6-007-signals-and-systems-spring-2011/pages/lecture-notes/)  [_](https://ocw.mit.edu/courses/res-6-007-signals-and-systems-spring-2011/pages/lecture-notes/)
 - B. A full download of the OCW course is on the [Box](https://ucmerced.box.com/s/5iv133o8k69fu96ddtj9xoqsf541azy)  [_](https://ucmerced.box.com/s/5iv133o8k69fu96ddtj9xoqsf541azy) folder for Fall 2026 EE 102
2. B. P. Lathi and Roger Green "Signal Processing and Linear Systems" Second Edition. ISBN: 9780190299040. E-book ISBN: 9780190857691
3. Signal Processing (with Python) by Vierinen and Jensen. Free textbook available [here](#) 

(https://github.com/jvierine/signal_processing_course/blob/main/signal_processing.pdf) and in class [Bo](https://ucmerced.box.com/s/kc06fnxrzw6133fa4ds74iljf771x3bi) (<https://ucmerced.box.com/s/kc06fnxrzw6133fa4ds74iljf771x3bi>) [here](https://ucmerced.box.com/s/5iv133o8k69fu96ddtj9xoqsf541azy) (<https://ucmerced.box.com/s/5iv133o8k69fu96ddtj9xoqsf541azy>) [x/Resources](https://ucmerced.box.com/s/kc06fnxrzw6133fa4ds74iljf771x3bi) (<https://ucmerced.box.com/s/kc06fnxrzw6133fa4ds74iljf771x3bi>)

4. Adams "Signals and Systems". Available [free](https://www.ece.uvic.ca/~frodo/sigsysbook/downloads/signals_and_systems-6.0.pdf) (https://www.ece.uvic.ca/~frodo/sigsysbook/downloads/signals_and_systems-6.0.pdf) and posted on [Box](https://ucmerced.box.com/s/5iv133o8k69fu96ddtj9xoqsf541azy) (<https://ucmerced.box.com/s/5iv133o8k69fu96ddtj9xoqsf541azy>)

Why take EE 102?

Signal processing and linear systems theory is foundational in engineering as it has revolutionized engineering in more ways than we realize (think: machine learning, AI, amplifiers, airplanes, medical devices, automotive vehicles, MRI scans, and pretty much every other engineering and science discipline out there). This is fantastic but there is also a downside! Learning signal processing is dependent on many prerequisites as it builds on various other fundamental courses in engineering. But at the same time, as electrical engineers, you are required to learn signal processing and linear systems theory. Other engineering disciplines typically do not require such a course (even though they should!) This gives you an edge because what you learn in this course is not just applicable to EE but also to all other areas. So, hopefully that is good enough motivation for you to cross the technical barriers (see prerequisites) in learning signal processing.

About EE 102 (catalog description)

Concepts and tools for continuous- and discrete-time signal and system analysis with applications in signal processing, communications, and control. Mathematical representation of signals and systems. Linearity and time invariance. System impulse and step responses. System frequency response. Frequency-domain representations: Fourier series and Fourier transforms. Introduction to linear systems and feedback. Time/frequency sampling and transfer functions. Continuous-discrete-time signal conversion and quantization.

Course Learning Outcomes

1. Students will learn to apply concepts from calculus to fundamental signals found in engineering systems. (links to PLO 1)
2. Students will learn to design and evaluate the properties of engineering systems for desired behavior. (links to PLO 6)
3. Students will be able to demonstrate the signal processing and linear systems concepts (CLO 1)

and CLO 2) in a computational project. (links to PLO 6 and 7)

Course prerequisites:

Circuit theory (ENGR 065), Linear Algebra and Differential Equations (MATH 024), and Vector Calculus (MATH 023) are all required prerequisites for this course. Strong mathematical preparation in calculus, vector algebra, complex numbers, linear algebra, and differential equations is desirable in this class. Further, examples from circuit theory will be used commonly as references and in examples. Finally, the concept of Laplace transforms from ENGR 065 will be useful in learning EE 102. Therefore, ENGR 065, MATH023, and MATH 024 are important and required pre-requisites for this course. In addition to math and circuits preparation, strong Python programming skills is also a pre-requisite!

Course policy and expectations:

1. **Lectures** will be designed as class discussions where you will be able to learn most of the key concepts required to complete the assignments successfully. Class activities will be designed to foster an enjoyable learning experience (rather than just passively listening to the instructor :). Attendance in lectures is required.
2. **Participation:** All in-class activities can be completed until midnight of the day of the class.
3. **Equipment:** You **must bring your laptop to the class**. Your laptop device will be useful to write and run code (refer to the [school suggested specifications \(https://it.ucmerced.edu/student-laptop-recommendation\)](https://it.ucmerced.edu/student-laptop-recommendation)) and work on various in-class activities. No other equipment is needed to learn signal processing.
4. **Labs** will be designed as **discussion sections**. **Attendance during lab sections is required** but you can **leave early** once you have finished the assigned tasks.
5. **Extensions:** For problem sets, you have **three free (no questions asked) extension requests**. Late work is still eligible for a partial credit: 7% deducted for every late day up to a maximum of 49%. So, if you submit a few minutes late, a 7% penalty for 1 day will be applied. That is, if your true score is 95, you will receive 88. If you submit 7 days late, you will receive ~half of the points you scored. That is, if your true score is 95 after submitting 7 days late, you will receive 48 points.
6. **Academic honesty:** All work you turn in must be your own (including the code). Collaborations are encouraged but the final work that you submit must be your own. Asking for help in understanding a concept, learning from discussions online, and asking ChatGPT for explanations are all OK — but, you must complete your own work after you have received this help and understood the concepts. If plagiarism is detected, you will be given a warning. On repeat offense, an action will be taken according to the university academic honesty policy [PDF \(https://osrr.ucmerced.edu/sites/osrr.ucmerced.edu/files/documents/academic_honesty_-_800.pdf\)](https://osrr.ucmerced.edu/sites/osrr.ucmerced.edu/files/documents/academic_honesty_-_800.pdf). More on

AI policy in EE 102 below:

1. **AI use for generating plotting code:** You are allowed to directly copy AI written code for plotting your results using matplotlib/MATLAB plot function.
2. **AI use for writing code that implements signal processing concepts:** Not allowed. You must write your own logic for the code. Your TA will ask you to explain your code logic during the discussion section (at random). We are not expecting you to demonstrate your mastery of programming syntax but your understanding of how the course concepts were implemented in code should be clear to you.
7. **Class Conduct and Community:** Remember that contributions from each of us can help in building a respectful, courteous, and an intellectually stimulating class environment. Language or behavior that prevents any student to participate fully in class is not acceptable. It is important to remain open to each other's thinking and engage in rigorous, challenging discussion about issues of shared concern. Please refer to the full student conduct policy [PDF \(https://osrr.ucmerced.edu/sites/osrr.ucmerced.edu/files/documents/code_of_conduct_600_and_700.pdf\)](https://osrr.ucmerced.edu/sites/osrr.ucmerced.edu/files/documents/code_of_conduct_600_and_700.pdf).
8. **Accommodations:** All accommodation requests are welcome. You can submit an accommodation letter through the Student Accessibility Services for this course. However, if you are unable to do so, you can directly discuss any accommodations that you may need with the instructor and we will do our best to provide a supportive learning environment for you.
9. **Support and Services for Undocumented Students:** In this time of uncertainty, the university has reaffirmed their commitment to supporting undocumented students. I would like to highlight these resources in the course syllabus to promote inclusion and belonging as we embark on this learning experience together, despite the uncertain and oppressive situation around us. At UC Merced, the Monarch Center offers support to undocumented and mixed-status students through resources, advocacy, and co-curricular programs aimed at promoting academic success and personal growth. Go here to learn more: <https://undoc.ucmerced.edu/> (<https://undoc.ucmerced.edu/>)
10. UC Merced is committed to providing an **equal opportunity environment** for all students and employees that remains free of all forms of discrimination, harassment, and exploitation. Discrimination and harassment based on race, color, religion, creed, sex, national origin, age, disability, veteran status, or sexual orientation is a violation of state and federal law and/or University policy and will not be tolerated. Retaliation against any person who complains about discrimination is also prohibited. If you witness or experience any form of harassment, please seek support and guidance. For more information, please visit <https://ophd.ucmerced.edu/> (<https://ophd.ucmerced.edu/>)
11. **Emailing/direct messaging on CatCourses:** If the syllabus does not answer your question, please reach out to the TA in the discussion section or the office hour. If your question needs urgent attention, you can send an email to the TA or the instructor (make your best judgement on who might answer your question best and quickest). Please **mention the course number in the subject** of any email you send.

Assessment structure:

1. In-class activities and quizzes can either be graded for participation or for correctness – **5% of your total grade.**
2. All reading assignments are graded for correct completion (you have unlimited attempts to do these). Two lowest scoring reading assignments will be dropped from final grade computation – **10% of your total grade.**
3. All problem sets require more independent effort than the reading assignments and are graded for correctness. Two lowest scoring problem sets will be dropped from the final grade computation – **20% of your total grade.**
4. Your summative assessment consists of four exams: three midterms and one final exam. Each midterm exam is worth 15% and the final exam is worth 20%. So, the exams contribute to **65% of your final grade.**

Grading Scheme

Letter Grade	Range
A+	100% to 99%
A	< 99% to 95%
A-	< 95% to 90%
B+	< 90% to 87%
B	< 87% to 83%
B-	< 83% to 80%
C+	< 80% to 77%
C	< 77% to 73%
C-	< 73% to 70%
D+	< 70% to 67%
D	< 67% to 63%
D-	< 63% to 60%
F	< 60% to 0%

Course timeline

Weekly Course Schedule

Date	Topics	Related assignments and due dates	Reading / Other notes
08/26	Week 0: Logistics, syllabus, and review of linear algebra	Pre-requisite #1 and #2 due on Sunday Aug 30	
08/31	Week 1 Lecture 1: Introduction to signals		
09/02	Week 1 Lecture 2: Properties of signals (including time invariance)	Pre-requisite #3 due on Sunday Sep 6 Discussion Pre-requisite #1: Vectors, projections, and linear algebra due Pre-requisite #2: Complex numbers due	
09/07	Week 2 Lecture 1: Labor Day holiday		----
09/09	Week 2 Lecture 2: The unit impulse and the unit step function	Discussion Pre-requisite #3: Circuit theory due Problem Set #1: Signals due	
09/14	Week 3 Lecture 1: System examples and expected outcomes		Last day to add/drop
09/16	Week 3 Lecture 2: Frequency modulation using complex exponentials and system properties	Discussion Problem Set #2: Properties of signals due	
09/21	Week 4 Lecture 1: Convolutions		
09/23	Week 4 Lecture 2: LTI systems, properties, stability, and applications and properties of convolution	Discussion Problem Set #3: LTI Systems due	
09/28	Week 5 Lecture 1: Fourier series: introduction, properties, and use		

Date	Topics	Related assignments and due dates	Reading / Other notes
	cases		
09/30	Week 5 Lecture 2: Midterm exam #1	Discussion Midterm exam #1 Problem Set #4: Convolutions due	Introductory signal analysis + initial Fourier topics
10/05	Week 6 Lecture 1: Applications of Fourier Series		
10/07	Week 6 Lecture 2: Fourier transform: introduction	Discussion Problem Set #5: Applications of convolution and its properties due	
10/12	Week 7 Lecture 1: Fourier transform applications		
10/14	Week 7 Lecture 2: Fourier transform applications	Discussion Problem Set #6: Fourier series due	
10/19	Week 8 Lecture 1: The frequency response		
10/21	Week 8 Lecture 2: Input-output analysis of systems	Discussion Problem Set #7: Applications of Fourier Series due	
10/26	Week 9 Lecture 1: Fourier Transform properties		
10/28	Week 9 Lecture 2: Midterm exam #2	Discussion Midterm exam #2 Problem Set #8: Practice midterm 2 and study guide due	Fourier analysis theory

Date	Topics	Related assignments and due dates	Reading / Other notes
11/02	Week 10 Lecture 1: Applications of Fourier analysis: EEG		
11/04	Week 10 Lecture 2: Applications of Fourier analysis: filtering	Discussion	Course withdrawal deadline 11/05
11/09	Week 11 Lecture 1: Sampling		
11/11	Week 11 Lecture 2: Veterans Day holiday	Problem Set #9: Audio password using DFT due	
11/16	Week 12 Lecture 1: Sampling: Nyquist theorem		
11/18	Week 12 Lecture 2: Midterm exam #3	Discussion Midterm exam #3 Problem Set #10: Filter design due	Filtering and sampling related topics
11/23	Week 13 Lecture 1: Applications of Fourier analysis: communication		Hybrid lecture option available.
11/25	Week 13 Lecture 2: Thanksgiving Holiday	Problem Set #11: Sampling due	
11/30	Week 14 Lecture 1: Guest lecture TBD Applications of filtering: image processing and 2D Fourier Transform		
12/02	Week 14 Lecture 2: Fast Fourier Transforms (FFT)	Discussion	
12/07	Week 15 Lecture 1: Review EE 102		
12/09	Week 15 Lecture 2: No class	Discussion No homework. Problem Set #12: Practice Final Exam due	Exam review